Thomas Puskas
Circuits 2 Lab Report Week 06
Due: 10-10-11
KUID: 2026962128

Experiment 04: Feedback

Objective:

Amplifiers are one of the most important configurations of transistors. The feedback system is divided into positive and negative feedback systems. One effect of the negative feedback system is that it reduces the gain, however the advantages far outweigh the gain loss. The advantages include gain desensitization, bandwidth extension, modifications of the circuits I/O impedance, and improvement of linearity. The purpose of this lab is to understand the uses and implementation of feedback systems. We also want to create a circuit and understand the effects of non-ideal impedances and the positive aspects of the negative feedback system.

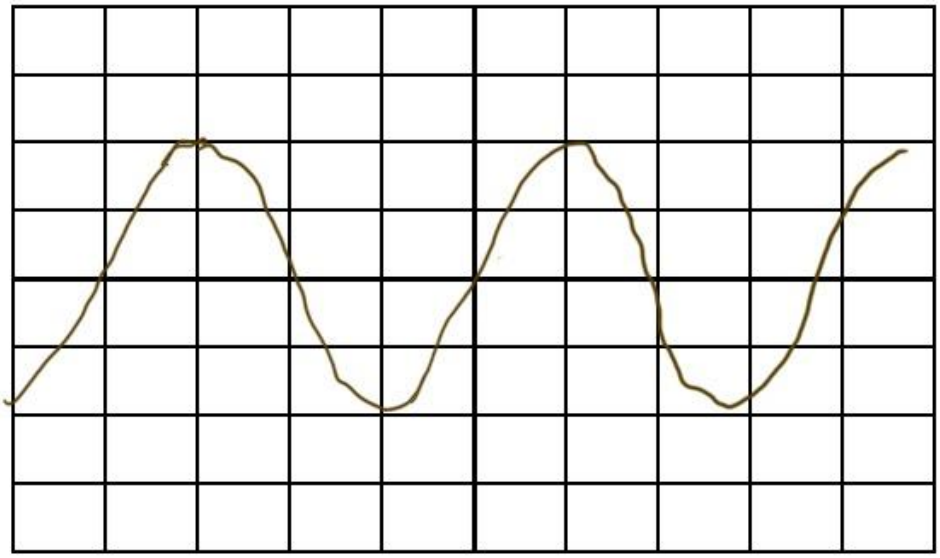
Materials:

Component		Quantity
Resistor	2.7 kΩ	1
	2.2 kΩ	1
	2 kΩ	1
	430 kΩ	1
	10 Ω	1
Capacitor	4.7 nF	1
NMOS	2N7000	2

3. Procedure

3.1 Properties of The Feedback System

2. Feedback (Closed-Loop) System



4.

Node A [Vpp]	Vout [Vpp]
0.368	0.926

5.

Vin [Vpp]	Closed [V/V]
0.01	92.8

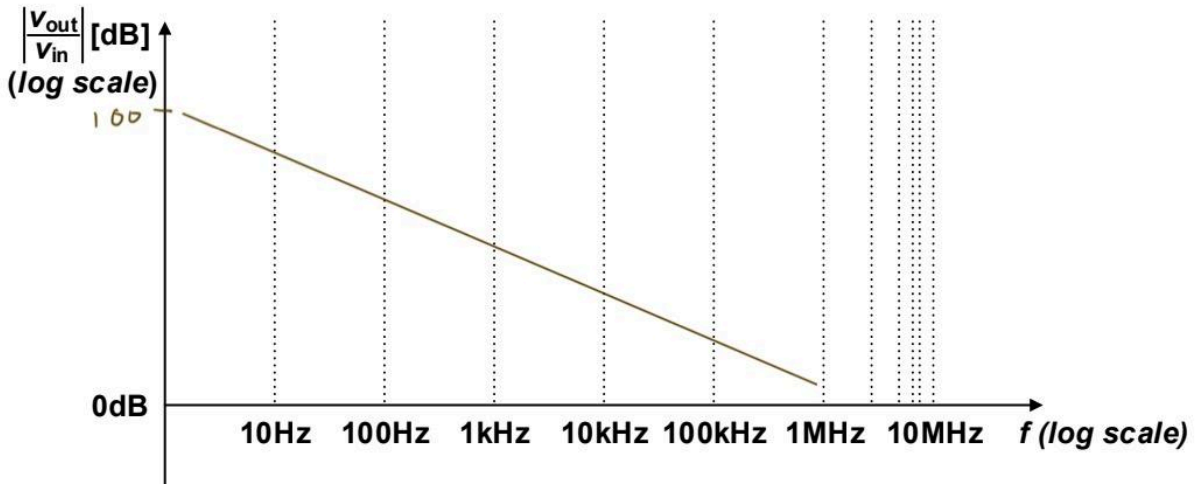
6.

Vin [V]	Node A [V]	Node B [V]	Node C [V]	Vout [V]
1.8726	2.4603	08824	0.110	8.721

8.

Frequency [Hz]	Vout [Vpp]	Closed [V/V]
10 Hz	0.328	32.8
100 Hz	0.832	83.2
1 kHz	0.848	84.8

10 kHz	0.872	87.2
20 kHz	0.873	87.3
30 kHz	0.792	79.2
40 kHz	0.736	73.6
50 kHz	0.736	73.6
100 kHz	0.672	67.2
200 kHz	0.568	56.8
300 kHz	0.408	40.8
400 kHz	0.360	36.0
500 kHz	0.336	33.6
1 MHz	0.192	19.2
2 MHz	0.216	21.6
3 MHz	0.152	15.2
4 MHz	0.136	13.6
5 MHz	0.136	13.6
f_{3-dB}	24.3 kHz	

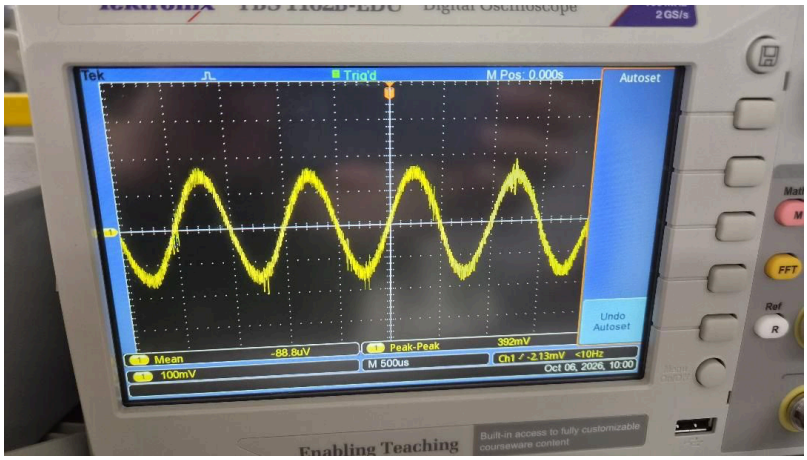


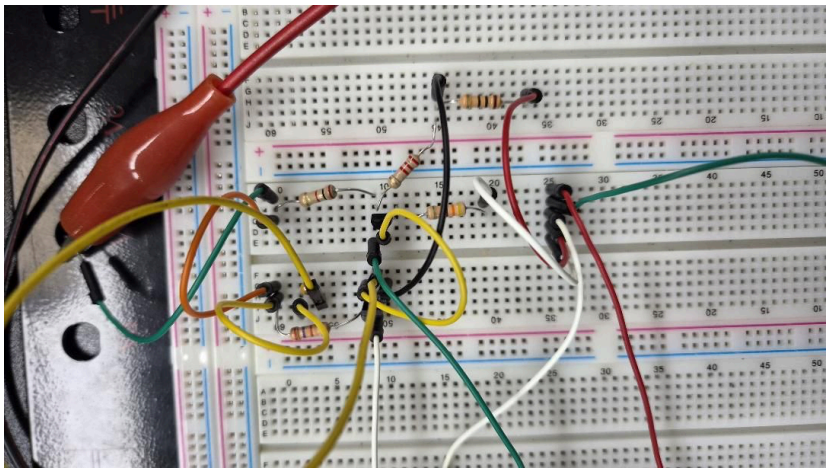
10.

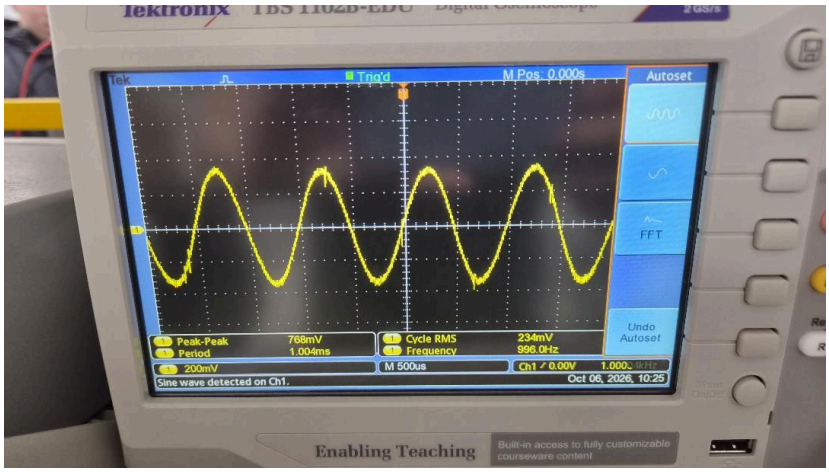
$V_{x,pp}$ [V_{pp}]	$i_{x,RMS}$ [mA_{rms}]	$i_{x,pp}$ [mA_{pp}]	$RV_{out,closed}$ [kΩ]
500 mV_{pp}	0.144	0.406	1.222
600 mV_{pp}	0.173	0.409	1.224
700 mV_{pp}	0.203	0.574	1.220
800 mV_{pp}	0.231	0.653	1.225
900 mV_{pp}	0.261	0.738	1.220
1000 mV_{pp}	0.356	0.979	1.021
Average of $RV_{out,closed}$			1.189

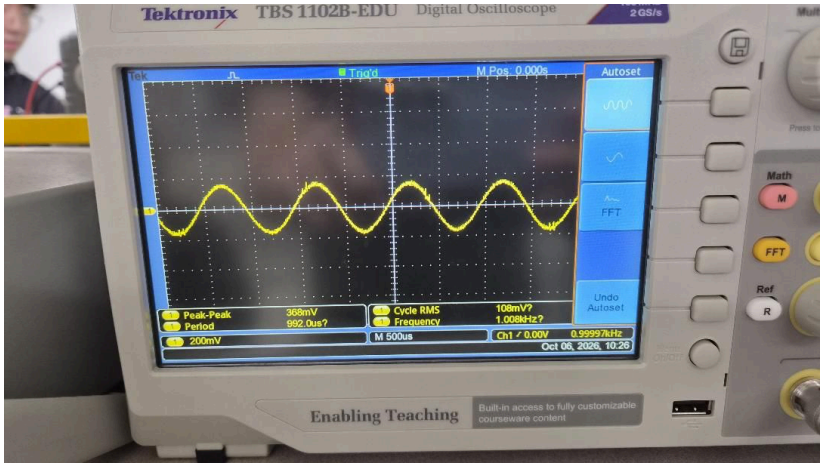
Photos

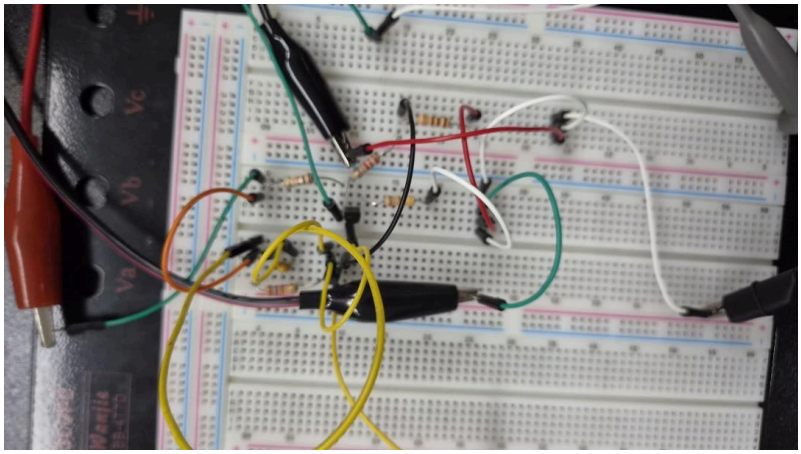


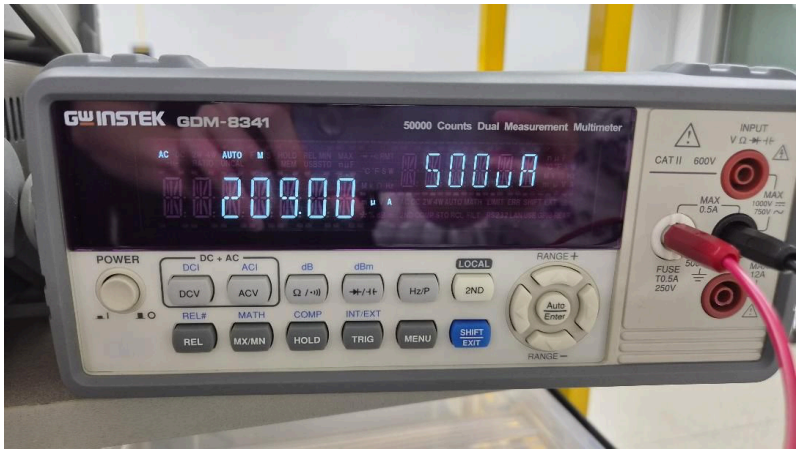





Tektronix
Tek
Trig'd
M Pos: 0.000s
Autoset
FFT
Undo
Autoset
Peak-Peak 768mV
Period 1.004ms
Cycle RMS 234mV
Frequency 996.0Hz
200mV
M 500us
Ch1 ∕ 0.00V
Sine wave detected on Ch1.
Oct 06, 2026, 10:25
Enabling Teaching
Built-in access to fully customizable courseware content

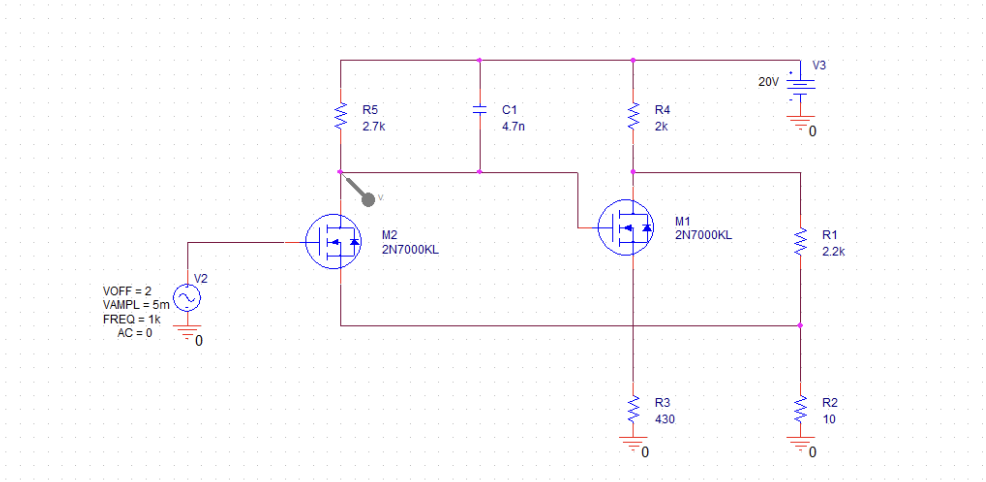

Tektronix
TBS 1102B-EDU
Digital Oscilloscope
2GS/s
Tek
Trig'd
M Pos: 0.000s
Autoset
FFT
Undo
Autoset
Peak-Peak 368mV
Period 992.0us?
Cycle RMS 108mV?
Frequency 1.008kHz?
200mV
M 500us
Ch1 ∕ 0.00V
0.99997kHz
Oct 06, 2026, 10:26
Math
M
FFT
Ref
R
Enabling Teaching
Built-in access to fully customizable courseware content

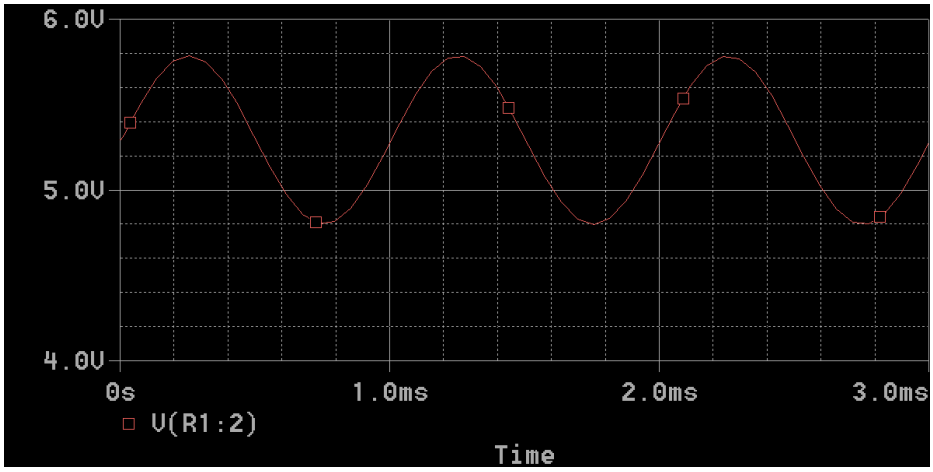


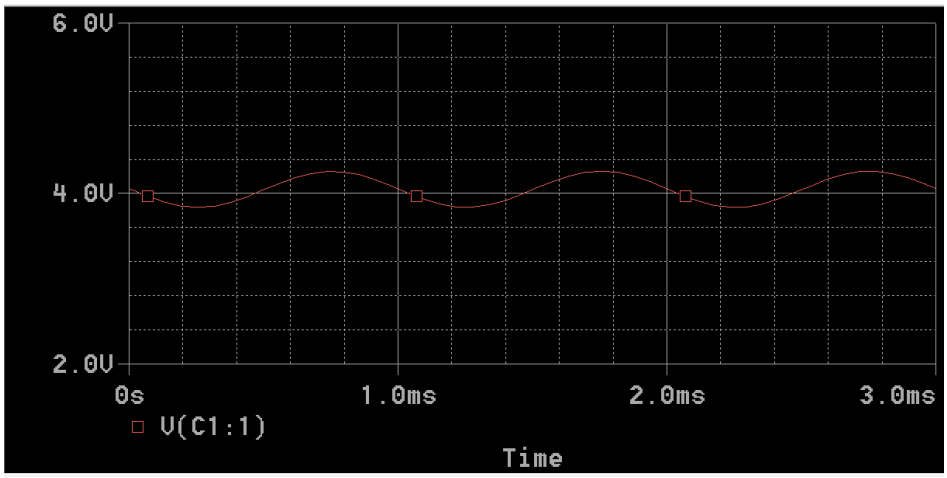

GW INSTEK GDM-8341
50000 Counts Dual Measurement Multimeter
209.00
500uA
POWER
DC + AC
DCV
ACV
REL
MX/MN
HOLD
TRIG
MENU
SHIFT EXIT
2ND
Auto Enter
INPUT
CAT II 600V


GW INSTEK AFG-2005
Arbitrary Function Generator
1.0000 kHz
FREQ
0.500 Vpp
9.00 V
OFST
FUNC
FREQ
AMPL
OFST
DUTY
Point
Value
ARB
Save/Recall
Hz/Vpp
kHz/Vrms
MHz/dBm

PSPICE:


R5
2.7k
C1
4.7n
R4
2k
20V
V3
0
M2
2N7000KL
M1
2N7000KL
R1
2.2k
V2
VOFF = 2
VAMPL = 5m
FREQ = 1k
AC = 0
0
R3
430
R2
10
0
0


6.0V
5.0V
4.0V
0s
1.0ms
2.0ms
3.0ms
V(R1:2)
Time


6.0V
4.0V
2.0V
0s
1.0ms
2.0ms
3.0ms
V(C1:1)
Time